Proceedings of IASS-IWSS 2026 – September 14-18, 2026, Turin, Italy
"R-Evolution of Shapes: Sustainability, Re-Use, and New Design Paradigms"
Annual Symposium of the International Association for Shell and Spatial Structures
3rd Italian Workshop on Shell and Spatial Structures
A. Manuello Bertetto, S. Gabriele, F. Marmo, A. Micheletti (eds.)

Topology-aware identification of independent edges in algebraic 3D graphic statics

Yao LU*, Yefan ZHI [a], Masoud AKBARZADEH [a,b]

* School of Architecture, University of Arizona, Tucson, USA
1040 N. Olive Rd, Tucson, AZ 85719
yaolu@arizona.edu

[a] Polyhedral Structures Laboratory, School of Design, University of Pennsylvania, Philadelphia, USA
[b] General Robotic, Automation, Sensing & Perception (GRASP) Lab, School of Engineering and Applied Science, University of Pennsylvania, Philadelphia, USA

Abstract

In three-dimensional graphic statics (3DGS), both form and force diagrams have geometric degrees of freedom (GDoF), meaning that edge lengths can vary while edge directions, reciprocity, and topology remain unchanged. GDoF is a whole number that indicates how many independent edge lengths must be specified to determine a unique diagram. Algebraic 3DGS (A3DGS) represents reciprocal diagrams in matrix form and enables the computation of GDoF through rank analysis. Previous approaches based on reduced row echelon form (RREF) can identify independent edges corresponding to these degrees of freedom. However, many valid edge sets can parameterize the same admissible solution space, and RREF typically produces only one edge basis that is often spatially sparse. This paper presents a method for selecting geometrically clustered edge bases for GDoF in reciprocal polyhedral diagrams to support more intuitive and controllable design exploration. The proposed framework computes the null space of the constrained algebraic system and applies QR decomposition with topology-aware pivoting to select independent edges that are not only algebraically valid, but also spatially clustered and easier to interpret as design controls. Each selected edge acts as an explicit geometric parameter whose length can vary independently while preserving equilibrium and reciprocity. In addition, the paper extends the discussion of GDoF from unconstrained reciprocal diagrams to constrained settings, including edge-length and vertex constraints, showing how constraints change both the number of GDoF and the corresponding edge bases. Several case studies demonstrate the effectiveness of the method. The framework has been implemented in the PolyFrame 2 plug-in for Rhino and Grasshopper, expanding the toolset for structural form-finding using algebraic 3D graphic statics.

Keywords: algebraic three-dimensional graphic statics, polyhedral reciprocal diagrams, geometric degrees of freedom, structural form-finding

1. Introduction

In three-dimensional graphic statics (3DGS) using polyhedral reciprocal diagrams, form and force diagrams have structural equilibrium encoded in their edge directions and topological reciprocity [1, 2]. The edge lengths can vary while preserving reciprocity, topology, and edge directions, resulting in geometric degrees of freedom (GDoF). The number of GDoF indicates how many independent edge lengths must be specified to determine a unique reciprocal diagram. While procedural and iterative methods can generate families of reciprocal diagrams from an input configuration, they do not explicitly compute the GDoF of the system. This may lead designers to unintentionally over-constrain reciprocal diagrams during form-finding by prescribing too many geometric parameters.

Copyright © 2026 by Yao LU, Yefan ZHI, and Masoud AKBARZADEH
Published by the International Association for Shell and Spatial Structures (IASS) with permission.

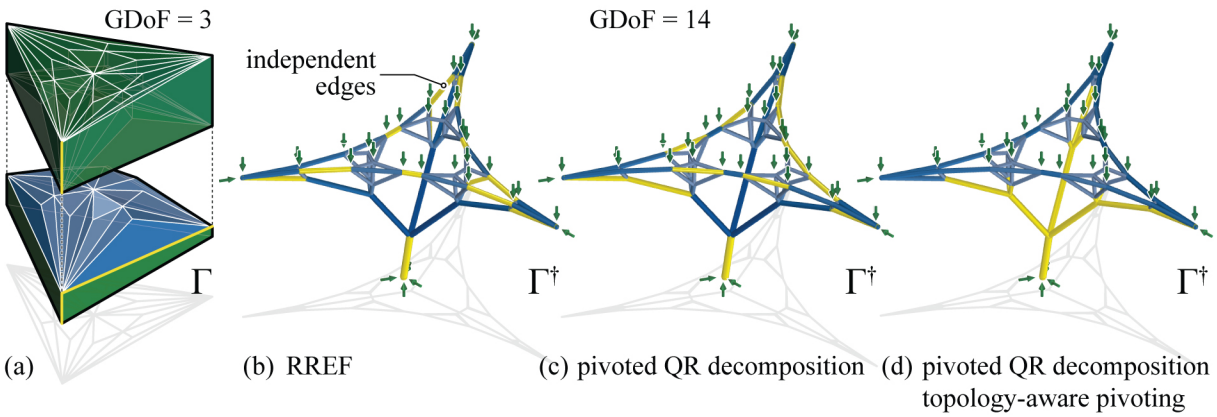


Figure 1: (a) Force diagram. Independent edge sets identified in the force diagram using (b) RREF, (c) pivoted QR decomposition, and (d) QR decomposition with topology-aware pivoting. The 14 selected independent edges are highlighted in yellow. The RREF and standard pivoted QR decomposition methods tend to select spatially sparse independent edges.

Algebraic three-dimensional graphic statics (A3DGS) represents reciprocal relationships through systems of linear equations and enables GDoF to be computed through rank analysis. Although the resulting GDoF value indicates the number of independent edge lengths in the reciprocal system, it does not directly reveal which geometric elements govern the admissible solution space. This limits the interpretability of algebraic solutions during structural form exploration.

Previous 2D and 3D algebraic approaches have identified independent edges corresponding to the GDoF through matrix reduction methods such as reduced row echelon form (RREF) [3–6]. While this provides a valid edge basis, it represents only one of many possible edge sets that can parameterize the same solution space. The resulting basis from the RREF method is often spatially sparse and geometrically scattered across the diagram, making the identified degrees of freedom difficult to use as intuitive design controls.

This paper presents a method for selecting geometrically clustered edge bases for GDoF in reciprocal polyhedral diagrams. Rather than treating independent edges as a fixed outcome of matrix reduction, the paper recognizes that multiple valid edge bases can represent the same admissible solution space. The proposed framework computes the null space of the constrained algebraic system and applies QR decomposition with topology-aware pivoting to select independent edges that are spatially clustered and easier to interpret as design controls (Figure 1). Each selected edge acts as an explicit geometric parameter whose length can vary independently while preserving equilibrium and reciprocity. In addition, the paper extends the discussion of GDoF from unconstrained reciprocal diagrams to constrained settings, including edge-length and vertex constraints, showing how constraints change both the number of GDoF and the corresponding edge bases. This paper focuses on the polyhedron-based formulation [1]; unless otherwise stated, the term 3DGS in this paper refers specifically to this polyhedron-based approach.

The feature has been implemented as a component in the PolyFrame 2 plug-in [7] for Rhino and Grasshopper. The paper is organized as follows. Section 2 reviews the background of algebraic three-dimensional graphic statics and geometric degrees of freedom. Section 3 presents the proposed method, including algebraic representation, rank-based GDoF computation, null-space analysis, and topology-aware pivoted QR decomposition. Section 4 demonstrates the method through several case studies. Section 5 discusses the implications and limitations of the approach, and Section 6 concludes the paper and outlines future work.

2. Background

2.1. Graphic statics and algebraic formulation

equilibrium is represented by closed force polygons at structural joints, with corresponding members shown in the form diagram [8–12]. The method has been used for both structural analysis and form-finding in built projects [13]. Its extension to three dimensions was first explored by Rankine and Maxwell [9, 14] and has recently been revitalized through computational methods. Two main technical trajectories have been developed: vector-based graphic statics (VGS) and polyhedron-based graphic statics (PGS). In VGS, the equilibrium of each node is represented by a closed cycle of 3D force vector [15]. In PGS, equilibrium is represented by closed polyhedra, where each face corresponds to an edge in the form diagram and the area of the face represents force magnitude [1, 2].

Both 2D and 3D graphic statics have procedural, iterative, and algebraic formulations. Procedural and iterative methods are useful for educational purposes because they allow diagrams to be constructed and adjusted intuitively. However, they can be relatively slow and may introduce numerical inaccuracies, especially when complex constraints are involved. Algebraic formulations address these limitations by expressing reciprocal relationships as systems of equations. In 2D, algebraic graphic statics has been developed from the algebraic duality of reciprocal diagrams [16] and later formalized through linear reciprocal relationships as an analysis tool [3]. This approach allows internal and external force magnitudes to be computed from a given form diagram and was later extended to a bidirectional interactive framework in which both form and force diagrams can be manipulated [4].

In 3D, Hablicsek et al. [5] established an algebraic framework in which reciprocal polyhedral diagrams are described by linear systems of equations enforcing closed coplanar edge vectors of polyhedral faces. This method can use either a form or force diagram as input and compute possible reciprocal diagrams through closed-form solutions with high accuracy. Later work expanded this algebraic framework to allow user-defined inputs for exploring the solution space [6], as well as quadratic formulations that accommodate prescribed face areas and edge lengths [17]. Lu et al. [18] further extended the method by incorporating edge-length and vertex constraints as additional constraint equations, improving the controllability of form and force diagrams during design adjustment. More recently, Lu [19] expanded the linear programming-based solving process to support objective-driven design exploration. Chai et al. [20] demonstrated the design potential of the algebraic framework by generating tension-compression forms from topologically complex, self-intersected force diagrams.

2.2. Geometric degrees of freedom (GDoF) and edge-based control

The term geometric degrees of freedom (GDoF) is closely related to the flexibility of reciprocal diagrams during form-finding and has been used in recent algebraic graphic statics research to describe the dimension of the admissible solution space of form and force diagrams [1, 5, 6, 18, 19]. Before the term was used, the concept of flexibility has been discussed in procedural and iterative approaches. However, these approaches cannot explicitly identify the dimension of the solution space or specify which edge lengths can serve as independent parameters.

Algebraic formulations make it possible to compute GDoF through rank analysis. This calculation provides the number of independent edge lengths in the form and force diagrams [5, 18]. Previous work for both 2D and 3D scenarios has also shown that independent edges can be identified through matrix reduction methods such as reduced row echelon form (RREF) [3–6]. In this way, GDoF can be connected to specific edges rather than treated only as an abstract numerical value.

However, the choice of independent edges is generally not unique. Multiple edge sets can parameterize the same solution space, and different selections may lead to different geometric interpretations. This observation provides the basis for the method developed in this paper, which focuses on selecting edge bases that are not only algebraically valid but also more suitable for geometric design control.

2.3. Problem Statement and objectives

A key gap remains between algebraic computation and geometric control in A3DGS. Existing methods can compute GDoF and identify a valid set of independent edges, but they do not guide which edge basis is most useful for design. This paper addresses this gap by selecting spatially coherent and interpretable edge bases for constrained reciprocal diagrams. The objective is to turn the algebraic solution space into explicit geometric controls, allowing designers to explore design variations more intuitively.

3. Method

The core contribution of this paper is a topology-aware pivoting strategy for identifying spatially clustered independent edges. In this paper, topology-aware pivoting refers to a modified QR pivoting process that uses adjacency and coplanarity weights to bias the selection of independent edges toward geometrically coherent regions of the diagram. It is built on column-pivoted QR decomposition, a rank-revealing matrix factorization. [21, 22]. Unlike RREF, which returns a single valid edge basis through matrix reduction, pivoted QR decomposition selects independent columns sequentially. This makes it possible to guide the selection process with additional geometric or topological criteria. While standard QR pivoting selects, at each step, the column with the largest residual norm, topology-aware pivoting adds preference scores for edges that are adjacent to, or coplanar with, previously selected pivots. This produces edge bases that remain valid for representing GDoF while being more clustered and easier to interpret as design controls.

This section is organized as three sub-sections. First, the constrained reciprocal diagram is formulated as a constrained algebraic system, where edge lengths are treated as unknown variables and edge-length or vertex constraints are incorporated into the matrix representation. Second, the null space of this system is used to define the admissible solution space, and QR decomposition with topology-aware pivoting is applied to select spatially clustered independent variables. Third, the selected algebraic variables are mapped back to their corresponding edges in the form or force diagram, allowing the GDoF to be visualized and manipulated as explicit geometric controls.

3.1. Algebraic representation of reciprocal diagrams with constraints

This subsection briefly summarizes the constrained algebraic formulation from previous work [18], as it provides the linear system from which GDoF and independent edge bases are extracted in this paper.

Let $\mathbf{q}$ denote the vector of unknown internal edge lengths in the form diagram. The closure of polyhedral faces is enforced by

$$\mathbf{Aq} = \mathbf{0},$$

where $\mathbf{A}$ is the closing matrix encoding the head-to-tail connectivity of edge vectors. Following the sign convention in [18], a positive value of q_i indicates compression, while a negative value indicates tension.

Prescribed edge lengths are represented as

$$\mathbf{Bq} = \mathbf{l},$$

where $\mathbf{B}$ is a selector matrix for constrained edges, and $\mathbf{l}$ stores their target lengths.

Vertex constraints prescribe relative positions between selected vertices and are written as

$$\mathbf{Dq} = \mathbf{d},$$

where $\mathbf{D}$ encodes filtered edge paths between constrained vertices, and $\mathbf{d}$ specifies their desired relative displacement.

The closure equations, edge-length constraints, and vertex constraints are assembled into one non-homogeneous linear system:

$$\mathbf{Mq} = \mathbf{t}, \tag{1}$$

with

$$\mathbf{M} = \begin{bmatrix} \mathbf{A} \\ \mathbf{B} \\ \mathbf{D} \end{bmatrix}, \qquad \mathbf{t} = \begin{bmatrix} \mathbf{0} \\ \mathbf{l} \\ \mathbf{d} \end{bmatrix}.$$

This constrained system defines the admissible solution space of the reciprocal diagram. The following steps use this system to compute GDoF and identify edge bases that can serve as geometric design controls.

3.2. QR decomposition with topology-aware pivoting

The term topology-aware is used here to describe pivoting guided by two diagram-based relationships: adjacency and coplanarity. The proposed method applies QR decomposition not to the original constraint matrix $\mathbf{M}$, but to the transpose of its null-space basis. This choice is directly tied to the rank calculation step. Since GDoF is computed through singular value decomposition (SVD) of $\mathbf{M}$, the null-space basis $\mathbf{N}$ is already available from the right singular vectors. Let e denote the number of unknown edge lengths and m denote the number of GDoF. When $m \ll e$, QR decomposition on $\mathbf{N}^T$ is computationally more efficient than applying QR decomposition to the full constraint matrix, because the pivoting step is performed on the reduced admissible variation space.

As a side note, the meaning of e depends on whether the diagram being solved is the form diagram or the force diagram. When solving for the form diagram, e corresponds to the internal face count of the force diagram. When solving for the force diagram, it corresponds to the edge count of the force diagram it self, following the convention by Lu et al. [18].

3.2.1. Rank and null space calculation

After the constrained system $\mathbf{Mq} = \mathbf{t}$ is assembled, its geometric degrees of freedom are computed from the rank of $\mathbf{M}$. Singular value decomposition (SVD) is used to determine the numerical rank and to extract a basis for the null space of $\mathbf{M}$. Let

$$\mathbf{M} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T,$$

and let r be the numerical rank of $\mathbf{M}$, determined by a tolerance on the singular values. For a system with e unknown edge lengths, the number of geometric degrees of freedom is denoted by m and is computed as

$$m = e - r.$$

The null-space basis is then obtained from the last m right singular vectors:

$$\mathbf{N} = \left[\mathbf{v}_{r+1}, \mathbf{v}_{r+2}, \ldots, \mathbf{v}_e\right], \qquad \mathbf{N} \in \mathbb{R}^{e \times m}.$$

3.2.2. QR decomposition on the null space

Once having the null space basis of $\mathbf{M}$, the solution $\mathbf{q}$ of Eq.1 can be written as

$$\mathbf{q} = \mathbf{q}_p + \mathbf{N}\boldsymbol{\alpha},$$

where $\boldsymbol{\alpha} \in \mathbb{R}^m$ is the vector of GDoF parameters, and $\mathbf{q}_p$ is any particular solution. Since $\mathbf{MN} = \mathbf{0}$, changing $\boldsymbol{\alpha}$ modifies edge lengths while preserving the same closing equations and geometric constraints.

Each row of $\mathbf{N}$ corresponds to one edge-length variable. Therefore, selecting m linearly independent rows of $\mathbf{N}$ is equivalent to selecting m independent edges that fully parameterize the admissible solution

space. To identify these independent rows, the method applies QR decomposition with column pivoting to the transpose of the null-space matrix:

$$\mathbf{N}^T\mathbf{P} = \mathbf{QR},$$

where $\mathbf{P}$ is a permutation matrix, $\mathbf{Q}$ is orthonormal, and $\mathbf{R}$ is upper triangular. Since $\mathbf{N}^T \in \mathbb{R}^{m\times e}$, the first m pivot columns selected by $\mathbf{P}$ correspond to m independent rows of $\mathbf{N}$, and therefore to m independent edges in the reciprocal diagram.

In standard column-pivoted QR, the pivot at step s is selected by maximizing the residual column norm:

$$p_s = \arg\max_{j\geq s} \|\mathbf{r}_j^{(s)}\|_2^2,$$

where $\mathbf{r}_j^{(s)}$ is the current residual of column j after previous Householder transformations. This rule identifies algebraically independent columns, but it does not consider the geometric distribution of the selected edges. As a result, the selected edge basis may be spatially scattered (Figure 1c).

3.2.3. Topology-aware pivoting with adjacency and coplanarity weights

For the force diagram, adjacency is defined between edges that share a vertex. For the form diagram, two forms of relationships are evaluated between force-diagram faces. The first is direct topological connectivity between edges. The second is coplanarity about faces. These relationships are used to bias the pivot selection toward geometrically clustered edge sets.

Each candidate edge j is assigned a topological weight w_j, initialized as

$$w_j = 1.$$

At each pivoting step, the selected column maximizes a weighted residual norm:

$$p_s = \arg\max_{j\geq s} \left(w_j \|\mathbf{r}_j^{(s)}\|_2^2 \right),$$

where $\mathbf{r}_j^{(s)}$ is the residual column after previous Householder transformations.

After a pivot edge p_s is selected, the weights of related edges are increased according to their geometric relationship with the selected pivot. Let $\mathcal{C}(p_s)$ denote the set of coplanar neighboring edges and $\mathcal{A}(p_s)$ denote the set of directly connected neighboring edges. The weight update is written as

$$w_j \leftarrow w_j + \beta_c, \qquad j \in \mathcal{C}(p_s),$$

and

$$w_j \leftarrow w_j + \beta_a, \qquad j \in \mathcal{A}(p_s),$$

where β_c and β_a are user-adjustable weight increments for coplanarity and topological adjacency, respectively. A larger β_c prioritizes coplanar selections, while a larger β_a prioritizes connected selections. Similar values balance planarity and connectivity. Figure 2 shows a branching tree example in which prioritizing adjacency produces a more intuitive edge basis, while Figure 3 shows prioritizing local planarity produces a more intuitive result.

To avoid selecting nearly dependent variables purely due to geometric weighting, the topological boost is applied only to columns whose residual norm remains above a numerical threshold. When topology-aware pivoting is disabled, all weights remain equal to one and the method reduces to standard column-pivoted QR decomposition.

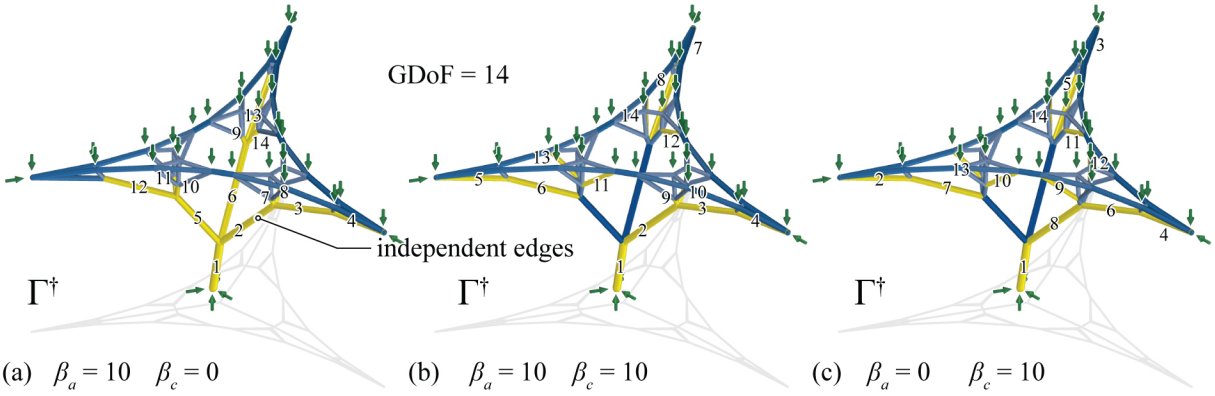


Figure 2: Different edge selections generated by varying the coplanarity weight increment (β_c) and adjacency weight increment (β_a). Numbers on the selected independent edges indicate the order of selection. (a) $\beta_c = 0$, prioritizing adjacency only. (b) Both β_c and β_a are nonzero, balancing adjacency and coplanarity. (c) $\beta_a = 0$, prioritizing coplanarity only.

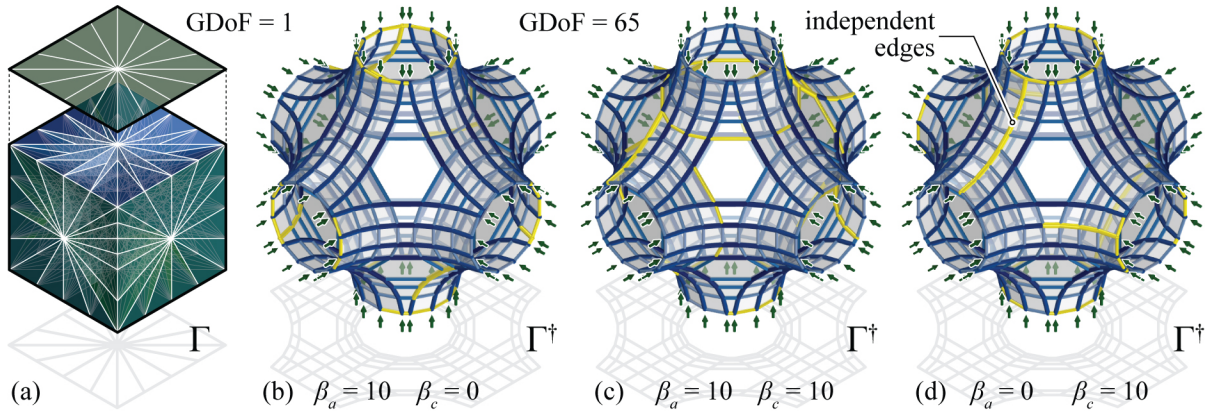


Figure 3: An example that shows prioritizing coplanarity produces a more intuitive selection of independent edges.

3.3. Mapping algebraic variables back to diagram edges

The first m pivot columns correspond to m independent rows of $\mathbf{N}$, and therefore to m independent edge-length variables. These pivot indices are mapped back to their associated edges in the form or force diagram.

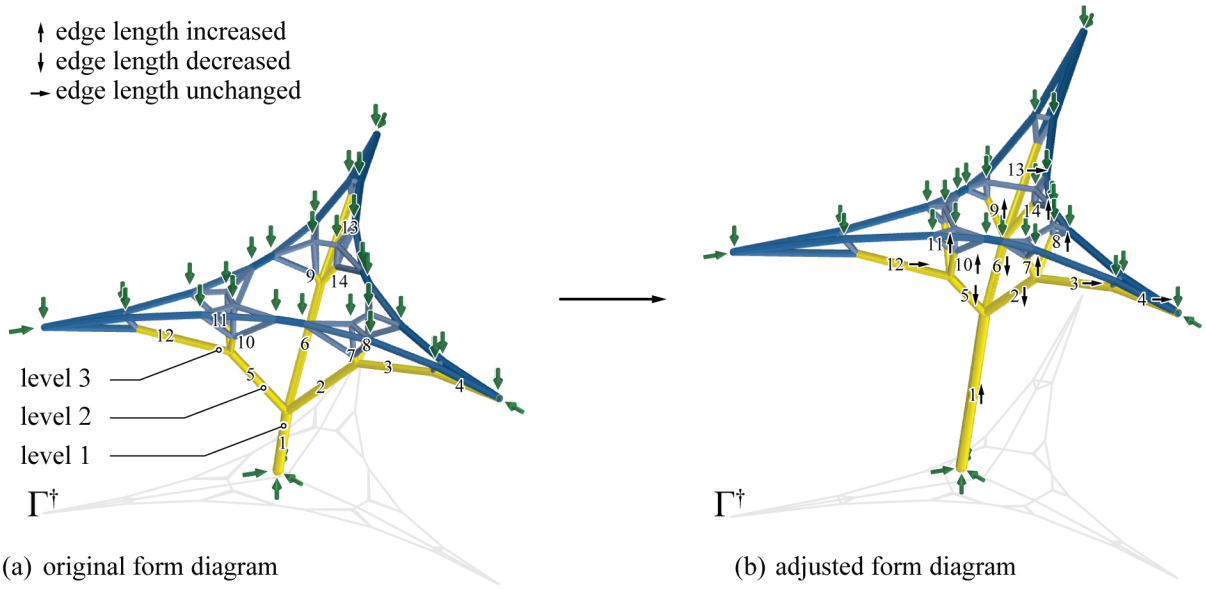
4. Case studies

4.1. Case study 1: design exploration guided by independent edges

The first case study uses independent edges to guide a branching form exploration. Starting from a root node, selected edge lengths are increased or decreased level by level to create a taller trunk and more condensed upper branches, while other edges remain unchanged. This interpretable edge set keeps the adjustable parameters manageable as shown in Figure 4.

4.2. Case study 2: GDoF under constraints

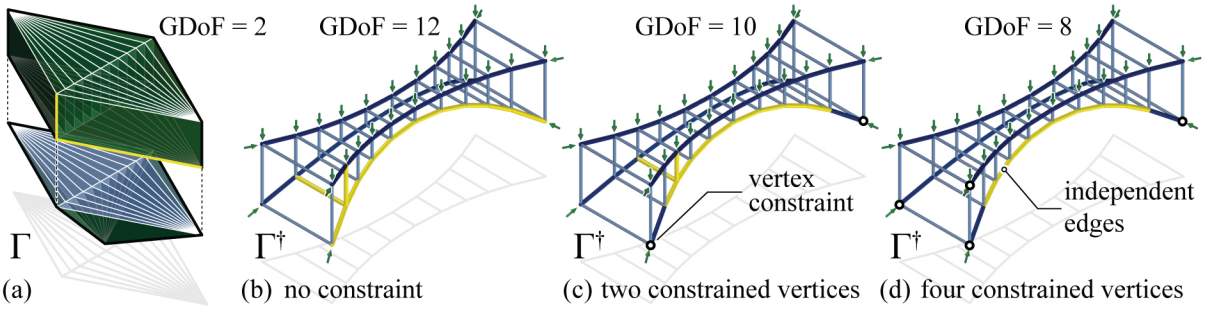
The second case study shows how vertex constraints reduce GDoF and update the corresponding independent edge basis. Starting from an unconstrained reciprocal diagram, additional vertex constraints are introduced, progressively reducing the admissible solution space. As shown in Figure 5, the GDoF decreases from the unconstrained case to the two-vertex and four-vertex constrained cases, demonstrating



(a) original form diagram (b) adjusted form diagram

Figure 4: Design adjustments guided by independent edges.

that GDoF depends on both diagram topology and imposed geometric constraints.



(a) (b) no constraint (c) two constrained vertices (d) four constrained vertices

Figure 5: Reduction of GDoF under increasing vertex constraints.

4.3. Case study 3: independent edges in complex diagrams

The third case study evaluates the method on more complex reciprocal form diagrams, where the readability of independent edge selections becomes more critical. Figure 6 compares edge bases identified by RREF, standard pivoted QR decomposition, and QR decomposition with topology-aware pivoting. All three methods produce valid edge sets for the same GDoF and show similar computational performance, with each completed in approximately 0.9 seconds on a desktop workstation. However, their spatial organization differs: RREF and standard pivoted QR tend to generate more scattered selections, while topology-aware pivoting produces a more coherent edge basis.

5. Discussion

Compared with previous methods using RREF and standard pivoted QR, topology-aware pivoting guides the selection toward clustered edges by prioritizing adjacency and coplanarity. RREF and standard pivoted QR remain valid for identifying independent edges, but their selections are driven mainly by largest algebraic independence or numerical residual strength. The resulting edge bases may be spatially scattered and less useful for direct manipulation.

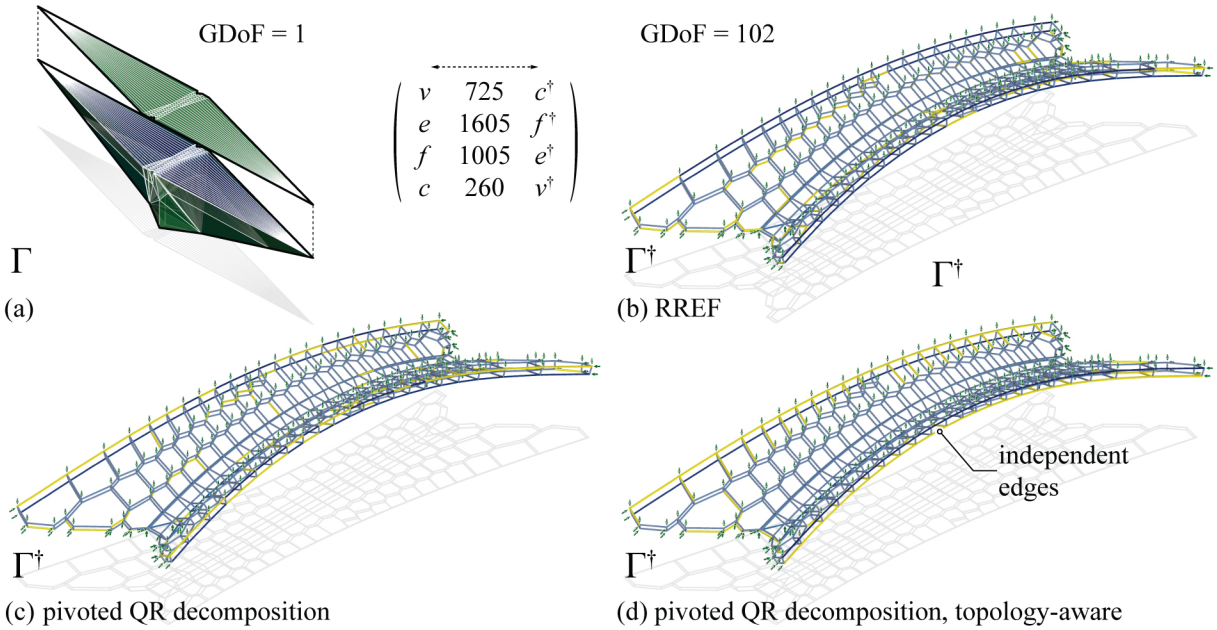


Figure 6: Independent edge selection in complex reciprocal diagrams using RREF, pivoted QR decomposition, and topology-aware pivoted QR decomposition.

It is important to note that the selected edge basis is not unique and depends on numerical tolerance, pivoting strategy, and weighting parameters. Future work could incorporate other selection criteria, such as symmetry, fabrication constraints, structural performance, or user-defined regions.

6. Conclusion

This paper presented a topology-aware QR decomposition-based method for selecting interpretable edge bases for GDoF in algebraic 3D graphic statics. The method computes the null space of a constrained reciprocal system, identifies independent edge variables through pivoted QR decomposition, and guides the selection using adjacency and coplanarity weights. Compared with RREF and standard pivoted QR, the proposed method preserves algebraic validity while producing edge sets that are more spatially coherent and easier to use as design controls. The method also helps designers visualize and understand how constraints affect both the number of available degrees of freedom and the edges that control them.

Acknowledgements

This research is funded by the Thomas Jefferson University Faculty Seed Grant awarded to Yao Lu.

References

[1] M. Akbarzadeh, "3d graphic statics using reciprocal polyhedral diagrams," Ph.D. dissertation, ETH Zurich, Zurich, Switzerland, 2016.

[2] J. Lee, "Computational design framework for 3d graphic statics," Doctoral Thesis, ETH Zurich, 2018, accepted: 2019-03-14T06:19:28Z. [Online]. Available: https://www.research-collection.ethz.ch/handle/20.500.11850/331210

[3] T. Van Mele and P. Block, "Algebraic graph statics," *Computer-Aided Design*, vol. 53, pp. 104 – 116, 2014. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0010448514000682

[4] V. Alic and D. Åkesson, "Bi-directional algebraic graphic statics," *Computer-Aided Design*,

vol. 93, pp. 26 – 37, 2017. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0010448517301446

[5] M. Hablicsek, M. Akbarzadeh, and Y. Guo, "Algebraic 3d graphic statics: Reciprocal constructions," *Computer-Aided Design*, vol. 108, pp. 30–41, 2019.

[6] M. Akbarzadeh and M. Hablicsek, "Geometric degrees of freedom and non-conventional spatial structural forms," in *Design Modelling Symposium Berlin*. Springer, 2019, pp. 3–17.

[7] PSL, "PolyFrame 2," https://www.food4rhino.com/app/polyframe-2, May 2023. [Online]. Available: https://www.food4rhino.com/app/polyframe-2

[8] W. J. M. Rankine, *A Manual of Applied Mechanics*. London: Griffin, 1858.

[9] ——, "Principle of the Equilibrium of Polyhedral Frames," *Philosophical Magazine Series 4*, vol. 27, no. 180, p. 92, 1864.

[10] J. C. Maxwell, "On Reciprocal Figures and Diagrams of Forces," *Philosophical Magazine Series 4*, vol. 27, no. 182, pp. 250–261, 1864.

[11] K. Culmann, *Die Graphische Statik*. Zürich: Verlag Meyer und Zeller, 1864.

[12] L. Cremona, *Graphical Statics: Two Treatises on the Graphical Calculus and Reciprocal Figures in Graphical Statics. Translated by Thomas Hudson Beare*. Oxford: Clarendon Press, 1890.

[13] D. P. Billington, *Robert Maillart's Bridges: The Art of Engineering*. Princeton University Press, 1979.

[14] J. C. Maxwell, "On reciprocal figures, frames and diagrams of forces," *Transactions of the Royal Society of Edinburgh*, vol. 26, no. 1, pp. 1–40, 1870.

[15] P. D'Acunto, J.-P. Jasienski, P. O. Ohlbrock, C. Fivet, J. Schwartz, and D. Zastavni, "Vector-based 3d graphic statics: A framework for the design of spatial structures based on the relation between form and forces," *International Journal of Solids and Structures*, vol. 167, pp. 58–70, Aug 2019.

[16] A. Micheletti, "On generalized reciprocal diagrams for self-stressed frameworks," *International Journal of Space Structures*, vol. 23, no. 3, pp. 153–166, Sep 2008.

[17] M. Akbarzadeh and M. Hablicsek, "Algebraic 3d graphic statics: Constrained areas," *Computer-Aided Design*, vol. 141, p. 103068, Dec 2021.

[18] Y. Lu, M. Hablicsek, and M. Akbarzadeh, "Algebraic 3d graphic statics with edge and vertex constraints: A comprehensive approach to extend the solution space for polyhedral form-finding," *Computer-Aided Design*, vol. 166, p. 103620, Jan. 2024.

[19] Y. Lu, "Objective-driven form-finding in algebraic 3D graphic statics via mixed-integer linear programming," in *Proceedings of IASS Symposium and Spatial Structures Conference 2025, The living past as a source of innovation*, México City, México, October 2025.

[20] H. Chai, Y. Lu, Y. Zhi, and M. Akbarzadeh, "Merged force diagrams for expanded tension-compression design space in polyhedral graphic statics," in *Advances in Architectural Geometry 2025*, C. Mueller, K. Gavriil, J. Ochsendorf, I. Fayyad, and J. Rabagliati, Eds., Cambridge, MA, US, Nov 16-19 2025.

[21] T. F. Chan, "Rank revealing qr factorizations," *Linear Algebra and Its Applications*, vol. 88–89, pp. 67–82, 1987.

[22] M. Gu and S. C. Eisenstat, "Efficient algorithms for computing a strong rank-revealing qr factorization," *SIAM Journal on Scientific Computing*, vol. 17, no. 4, pp. 848–869, 1996.